A Multistage Semi-Supervised Network for Hyperspectral Super-Resolution

Xiaotong Qi, Yang Xu, Ke Zheng, Jiaxin Li, *Student Member, IEEE*, Le Yu, Yuhang Zhao, and Chengyue Hu

***Abstract*— Hyperspectral imaging can capture abundant spectral information and reveal the spectral absorption properties of surface materials. Nevertheless, the tradeoff in spatial resolution reduces its capacity to represent surface object textures and structures; hyperspectral super-resolution (SR) technology is a viable solution to this problem. Yet, mainstream supervised methods depend on low-scale training data and specific data distributions, restricting their generalization capability and practicality in real scenarios. Although unsupervised methods remove the reliance on training data, they still face suboptimal reconstruction quality due to the absence of reference images and inaccuracies in degradation process estimation. Furthermore, bridging the performance gap between simulated datasets and real-world applications remains challenging. To this end, we propose a semi-supervised network that effectively couples unsupervised learning and supervised pretraining in multistage architecture, MCS-Net for short. The network consists of three key components: degradation information estimation (DIE), supervised fusion pretraining (SFP) at low resolution, and unsupervised image generation (UIG) at full resolution. The MCS-Net first estimates deep degradation information from input image pairs using DIE. It then applies supervised learning in SFP to construct a pretrained fusion function and its parameters from the input low-resolution data pairs and their fused outputs. Finally, the pretrained parameters from the previous stage are used to initialize the fusion network of UIG, which is then fine-tuned under the guidance of degradation parameters estimated by DIE, enabling the network to process full-resolution images effectively. Ablation experiments validated the effectiveness of each component. Moreover, the proposed MCS-Net outperforms the existing state of the art (SOTA) methods across six evaluation metrics in the simulation experiments, and the experimental results on real satellite data further validate the outstanding image fusion performance of MCS-Net and its potential for practical applications.**

***Index Terms*— Hyperspectral image (HSI) classification, semi-supervised learning, super-resolution (SR).**

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I. Introduction

HYPERSPECTRAL imaging technology often captures tens–hundreds of narrow and contiguous spectral bands across the spectral range with fine sampling intervals, delivering abundant spatial, spectral, and radiometric information for Earth's surface observations. It is recognized as a pivotal technology in the domain of remote sensing [1], [2]. Hyperspectral images (HSIs) obtained through this technology are widely used in various industries. Moreover, with its superior spectral resolution over broadband satellite images, HSI has found broad applications in data processing domains such as target detection [3], [4], [5], land cover classification [6], [7], [8], and multimodal data fusion [9], [10], [11]. However, hyperspectral imaging systems face a tradeoff between spatial and spectral resolutions, with coarse spatial details significantly limiting the scope of HSI applications. Meanwhile, multispectral images (MSIs) exhibit complementary characteristics, offering high spatial resolution but with relatively limited spectral information. These two heterogeneous images each possess unique yet complementary properties [12].

Early studies applied pan-sharpening methods to hyperspctral image super-resolution (HSISR), including component substitution (CS) and multiresolution analysis (MRA). These two strategies can be extracting detailed information from high-resolution images and injecting it into low-resolution images, collectively called detail injection methods. However, these methods often suffer from significant spatial–spectral distortion problems [13], [14]. Unlike the direct strategies of detail injection methods, model optimization-based approaches account for more complex degradation relationships, such as Bayesian representation, matrix decomposition, and tensor representation [15]. These methods address the ill-posed inverse problem of fusion by designing appropriate regularization terms. While they are theoretically interpretable and feature unsupervised characteristics (not relying on reference images), the challenge lies in the difficulty of manually designing precise priors and sensor parameters, which limits their practicality [16].

Deep learning (DL) possesses a powerful capability to model arbitrary nonlinear relationships between data, opening

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new avenues for HSISR research [17], [18], [19], [20]. Specifically, most existing studies adopt supervised learning strategies, relying on downsampling an image to obtain low-scale simulated images and using the original image as a reference high-resolution HSI (HrHSI) to construct the training dataset [21]. Leveraging these triplet datasets, the network learns the mapping relationship from low-resolution to high-resolution images and applies the well-trained model to test images to obtain HrHSI. While supervised methods can efficiently enhance low-resolution HSI's (LrHSI's) spatial resolution under specific training conditions, relying on low-scale simulated data for training distorts the original data features, and in real-world scenarios, HrHSI ground truth (GT) is often unavailable, posing challenges for the practical application of supervised learning methods. Additionally, the superior performance of supervised networks is confined to specific imaging models, with their performance degrading significantly when test data originate from varied sensors or observation scenarios [22]. Conversely, unsupervised networks require only a pair of HSI–MSI data as input to adaptively learn the underlying priors of the input data at full-resolution scale and construct loss functions through a degradation model, thereby effectively addressing the fusion inverse problem [23], [24], [25]. Nevertheless, unsupervised methods frequently suffer from unstable performance caused by the lack of reference images and imprecise degradation process estimation. Hence, improving unsupervised methods' robustness and generalization capacity under conditions lacking accurate reference data remains a pressing issue. Moreover, the high-dimensional nature of HSISR tasks results in elevated computational complexity for certain methods, further constraining their practical efficiency. On the other hand, image fusion aims to facilitate downstream applications, necessitating the resolution of performance gaps in fusion models between simulated datasets and real-world applications. However, current studies fail to adequately verify model performance and usability in real-world satellite data.

Fig. 1 illustrates the fusion result differences of the aforementioned traditional, deep supervised, and deep unsupervised methods, highlighting each approach's challenges. Here, Fig. 1(b)–(d) shows the results of coupled nonnegative matrix factorization (CNMF) [26], SSR-Net [18], and UMC2FF [27], respectively, which represent the three method categories for comparison. It can be observed that the fusion results of CNMF and UMC2FF exhibit significant color variation compared to the GT [as shown in Fig. 1(a)]. The former is primarily due to the limited expressiveness of the designed priors, while the latter arises from the lack of reconstruction references. Although the supervised method SSR-Net achieves superior visual effects, its dependence on triplet training data undermines its practicality and insufficiently uses the original information. Therefore, combining the advantages of both DL paradigms in the HSISR task remains an urgent research challenge.

Fortunately, related studies have successfully applied self-supervised, semi-supervised methods, and zero-shot learning [28], [29], [30], [31], [32] in natural image super-resolution (SR) and remote sensing pan-sharpening to address the challenges of limited training data and dependence, achieving

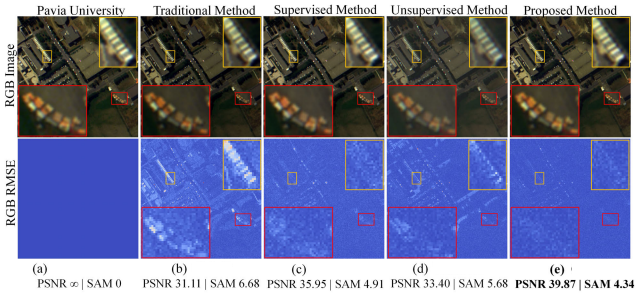


Fig. 1. Visualization of fusion results from different methods for the HSISR task demonstrates the differences in both visual effects and quantitative measurements. Among them, CNMF [26] is the traditional method, SSR-Net [18] is the supervised method, and UMC2FF [27] is the unsupervised method. (a) GT. (b) CNMF. (c) SSR-Net. (d) UMC2FF. (e) Ours.

improved SR performance. In this article, we propose a semi-supervised network for HSISR that effectively couples unsupervised learning and supervised pretraining in multistage architecture, termed MCS-Net. The network consists of three components: degradation information estimation (DIE), supervised fusion pretraining (SFP) at low resolution, and unsupervised image generation (UIG) at full resolution. The DIE component enables our network to perform reconstruction in an unknown SR manner. The network adaptively learns the spectral response function (SRF) and point spread function (PSF) without needing prior knowledge. Subsequently, we designed a residual spectral calibration fusion network (RSCF-Net) for supervised pretraining in the SFP component. By embedding and integrating residual structures and spectral attention modules within the network, RSCF-Net effectively performs adaptive calibration of key spectral responses while extracting and reusing essential features. Finally, the UIG component initializes with the pretrained parameter from SFP and fine-tunes the full-resolution fusion network under the guidance of degradation parameters estimated by DIE. The main contributions of this study are summarized as follows.

1) Inspired by the successful application of semi-supervised learning in natural image SR and pan-sharpening, we propose a novel semi-supervised network for the HSISR task to address the challenges faced by existing image fusion methods.
2) The proposed MCS-Net successfully integrates the advantages of both unsupervised and supervised learning paradigms, completing HSISR without additional training data or precise prior knowledge, and offers new potential for enhancing fusion performance in real-world scenarios.
3) The proposed semi-supervised framework exhibits excellent flexibility and scalability, enabling integration with existing DL architectures, and significantly improves the model's SR performance.

The structure of this article is as follows. Section II reviews the development of HSISR. Section III details the architecture and formulation of MCS-Net. Section IV presents the experimental results and analysis on simulated and real datasets, additionally verifying the framework's compatibility. Section V concludes this study.

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II. Related Work

In this section, the primary methods for HSISR tasks are categorized into traditional and DL approaches for a concise overview.

A. Traditional Methods

Among the traditional methods based on pan-sharpening, Gomez et al. [33] successfully applied wavelet technology to HSISR by generating fusion images through a layer-by-layer combination of wavelet coefficients. Zhang and He [34] utilized 3-D wavelet transforms to process spatial and spectral information simultaneously, achieving more efficient feature integration. Selva et al. [35] proposed a hypersharpening framework capable of synthesizing high-resolution images for each hyperspectral band. However, due to the limited capacity of these methods for detailed extraction and injection, they frequently suffer from noticeable spatial and spectral distortions.

Another class of traditional methods reconstructs the target image by introducing appropriate priors and sensor parameters as regularization terms and solving optimization problems. These methods include Bayesian methods, matrix decomposition, and tensor representation. In the first paradigm, Simões et al. [36] proposed the HySure algorithm based on subspace regularization and treats data fusion as a convex optimization problem. Wei et al. [37] developed the fast fusion based on Sylvester equation (FUSE) algorithm based on the Sylvester equation, which calculates Bayesian estimates under different priors, significantly reducing computational complexity. However, when the image content is complex, these Bayesian-based methods fail to produce satisfactory fusion results. In another research direction, matrix decomposition methods decompose the 3-D target image into solving 2-D abundance and endmember matrices. Yokoya et al. [26] proposed an unsupervised unmixing-based CNMF algorithm that alternately unmixes hyperspectral and multispectral data to generate endmember and abundance matrices. Akhtar et al. [38] incorporated spatial structure and matrix nonnegativity to constrain the sparse optimization process. However, matrix decomposition methods often disrupt the internal structure of the data, making it difficult to effectively utilize the latent information of the original cube data. In contrast, tensor representation methods preserve the inherent structure of the original data. Li et al. [39] decomposed HSI into a sparse core tensor product of three dictionaries and proposed a coupled sparse tensor decomposition-based approach. Xu et al. [40] imposed sparsity constraints on the core tensor and unidirectional total variation on three dictionaries to maintain image sparsity and piecewise smoothness. Such methods, which regard the target HrHSI as comprising a core tensor and three dictionaries, are generally known as Tucker decomposition. Additionally, Xu et al. [41] constructed a nonlocal similar patch tensor of the image and applied coupled canonical polyadic (CP) tensor decomposition to share factor matrices. Ding et al. [42] applied block term decomposition (BTD) to HSISR for the first time, using a block decomposition tensor model with physical interpretation to ensure SR image recoverability under low-rank constraint conditions.

B. DL-Based Methods

DL-based HSISR methods achieve outstanding reconstruction performance by leveraging convolutional neural networks' powerful intrinsic feature learning capabilities. According to the fusion paradigm, HSISR methods are generally classified into supervised and unsupervised learning approaches. The supervised learning method trains the network using additional triplet data to obtain specific priors for the research scene under different observation modes, and these priors are then applied to the test images to accomplish the SR task. Wang et al. [43] introduced a deep convolutional neural network to learn the end-to-end mapping between image pairs, integrating a spectral loss function to guarantee precise spectral reconstruction. Considering the cubic structure of HSI data, Palsson et al. [17] proposed a 3-D convolutional neural network for HSI fusion, aiming to strengthen the model's feature extraction capability. In addition to single-stream networks, Yang et al. [44] designed a dual-branch fusion framework, with one branch dedicated to reconstructing the target image and the other leveraging an attention network to extract spatial high-frequency details. Zhang et al. [18] improved the edge details and fidelity of the reconstructed target by introducing a spatial–spectral edge loss after obtaining the preliminary fusion results from the network. Hu et al. [45] proposed a network leveraging attention mechanisms to retain spatial and spectral information, enhancing generalization and reducing computational complexity. Wang et al. [46] introduced a U-Net architecture integrating joint spatial–spectral attention mechanisms and full dense connections, facilitating flexible combinations of multiscale and multipath features, but the cost of increased computational load. Incorporating other DL architectural aspects, Xiao et al. [47] employed a generative adversarial network (GAN) for fusion, embedding a degradation model between the generator and discriminator and optimizing it using a combined adversarial and $L1$ loss function. Jia et al. [48] devised an innovative multiscale spatial–spectral Transformer network. With research progress, supervised methods have demonstrated increasingly outstanding fusion performance. However, supervised methods still face issues such as dependence on external training data, insufficient utilization of raw information, and poor generalization of specific priors, which significantly limit their practical applicability in real-world scenarios.

To mitigate the reliance on training datasets and GT images, researchers have focused on applying unsupervised approaches to HSISR tasks. In the early, Qu et al. [49] pioneered an unsupervised encoder–decoder network to address the limitations of supervised fusion methods, effectively reducing spectral distortion by minimizing the angular differences between the two modality representations. Inspired by the unsupervised characteristics of traditional methods, Zheng et al. [50], Yao et al. [51], Liu et al. [52], and Li et al. [53] applied matrix factorization techniques, combining deep networks to extract the potential abundances and corresponding endmembers from the input modalities, and then reconstruct HrHSI. In recent studies, Cao et al. [25] proposed a multilevel Transformer network to utilize hierarchical features effectively. Dong et al. [54]

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combined the denoising diffusion probabilistic model with the principles of HSISR, progressively reconstructing by learning its distribution at different scales. To address the error accumulation caused by misregistration, Qu et al. [55] proposed a progressive multi-iteration registration-fusion collaborative optimization network for unregistered images. Zhang et al. [56] leveraged a pretrained diffusion model to learn image prior information, promoting high-quality reconstruction of a Transformer-based SR model. Additionally, Li et al. [57], [58] developed a multistage learning framework that extracts image priors from latent model information, enhancing the reconstruction potential using deep image prior (DIP). Although unsupervised methods perform excellently in addressing the significant limitations of the supervised paradigm, they often suffer from severe fluctuations in generalization performance. Therefore, it becomes crucial to explore and develop a method that combines the advantages of both paradigms [28], [29], [30].

III. Proposed Method

A. Overall Framework

The HSISR task aims to enhance the spatial resolution of LrHSI by fusing HrMSI from the same observed scene, thereby estimating the potential HrHSI. Although deep supervised methods exhibit excellent reconstruction performance, this article remains dependent on training datasets and fails to overcome this limitation. In contrast, deep unsupervised methods can fully utilize raw information without relying on any training data, offering better SR performance compared to traditional unsupervised methods and enabling their application in real-world HSISR tasks. However, their reconstruction capability is often limited due to the lack of accurate reconstruction references. Therefore, we propose a multistage semi-supervised image fusion network, MCS-Net, to combine the advantages of both DL strategies. It consists of three distinct components, each optimized at its respective stage. Specifically, the DIE component adaptively estimates degradation information from observed LrHSI–HrMSI pairs using a micronetwork, including PSF, SRF parameters, and LrMSIs derived from the two cross-modal data, aiming to address the challenge of obtaining degradation priors in real-world applications. Subsequently, we designed an RSCF-Net for the SFP component to exploit and integrate spatial and spectral information better. This component performs low-resolution SFP with explicit reference data. Finally, the pretrained parameter from SFP is used to initialize the UIG's fusion network, which is trained under the guidance of degradation parameters estimated by the DIE component, facilitating effective fine-tuning at full resolution. Fig. 2 presents the overall architecture of the proposed MCS-Net. A detailed explanation of each component is provided in the following.

B. Degradation Information Estimation

We unfold the 3-D tensor data and represent HrMSI as $\mathbf{Z} \in \mathbb{R}^{HW\times c}$, LrHSI as $\mathbf{Y} \in \mathbb{R}^{hw\times C}$, and HrHSI as $\mathbf{X} \in \mathbb{R}^{HW\times C}$, where H/h, W/w, and C/c represent the height, width, and the number of spectral bands of the corresponding images, respectively. Specifically, $H \gg h$, $W \gg w$, and $C \gg c$. The observed $\mathbf{Y}$ and $\mathbf{Z}$ can be related to the unknown $\mathbf{X}$ through a degradation model, which describes the spatial and spectral downsampling processes

$$\mathbf{Y} = \mathbf{PX} \tag{1}$$

$$\mathbf{Z} = \mathbf{XS} \tag{2}$$

where $\mathbf{P} \in \mathbb{R}^{hw\times HW}$ represents the PSF for spatial degradation matrix, and $\mathbf{S} \in \mathbb{R}^{C\times c}$ represents the SRF for spectral degradation matrix.

By applying the spatial downsampling operator $\mathbf{P}$ from (1) to $\mathbf{Z}$ and the spectral degradation operator $\mathbf{S}$ from (2) to $\mathbf{Y}$, it can be concluded that the low-resolution MSI (LrMSI) $\mathbf{L}_1$ obtained from the spatial downsampling of Z should approximate to the LrMSI $\mathbf{L}_2$ obtained from the spectral downsampling of Y. The derivation process is presented as follows:

$$\mathbf{L}_1 = \mathbf{YS} \tag{3}$$

$$\mathbf{L}_2 = \mathbf{PZ} \tag{4}$$

substituting (2) and (1) into (3) and (4) allows us to infer that

$$\mathbf{L}_1 = \mathbf{L}_2 = \mathbf{PXS} \tag{5}$$

where $\mathbf{L}_1$ and $\mathbf{L}_2 \in \mathbb{R}^{hw\times c}$.

According to (5), potential degradation information can be estimated from the input image pairs by designing a suitable network structure to learn the spatial and spectral degradation processes. Specifically, DIE is a parallel yet noncoupled dual-stream convolutional neural network that estimates $\mathbf{P}$ and $\mathbf{S}$, which are used for spatial and spectral degradations in (5) through convolution operations. For the spatial degradation learning stream, a shared convolution kernel with a stride of $r \times r$ is applied to independently convolve each spectral band, where r is the ratio factor of the input image's ground sampling distance (GSD), and $\mathbf{P}$ is updated iteratively as a trainable parameter in the network. Similarly, the spectral degradation learning stream is implemented by a 1×1 convolution layer, where $\mathbf{S}$ is the corresponding kernel parameter. Clamp activation and normalization operators constrain the trainable parameters $\mathbf{P}$ and $\mathbf{S}$, ensuring that the estimated PSF and SRF parameters satisfy nonnegativity and the sum-to-one constraint. Given the input LrHSI $\mathbf{Y}$ and HrMSI $\mathbf{Z}$, their degradation modeling process through DIE can be formulated as

$$\mathbf{Z}_{\mathrm{lr}}^2 = \mathcal{F}_z(\mathbf{Z}, \boldsymbol{\theta}_{\mathrm{PSF}}) \tag{6}$$

$$\mathbf{Z}_{\mathrm{lr}}^1 = \mathcal{F}_y(\mathbf{Y}, \boldsymbol{\theta}_{\mathrm{SRF}}) \tag{7}$$

where $\mathbf{Z}_{\mathrm{lr}}^1$ and $\mathbf{Z}_{\mathrm{lr}}^2$ represent two LrMSIs. $\mathcal{F}_z(\cdot, \boldsymbol{\theta}_{\mathrm{PSF}})$ denotes the spatial degradation learning stream in DIE, and $\boldsymbol{\theta}_{\mathrm{PSF}}$ is its weight parameter, while $\mathcal{F}_y(\cdot, \boldsymbol{\theta}_{\mathrm{SRF}})$ is the spectral degradation learning stream in DIE, and $\boldsymbol{\theta}_{\mathrm{SRF}}$ is the parameterized representation of SRF. The optimization process of network training can be formalized as an energy-minimization problem

$$\boldsymbol{\theta}_{\mathrm{PSF}}^*, \boldsymbol{\theta}_{\mathrm{SRF}}^* = \min_{\boldsymbol{\theta}_{\mathrm{PSF}}, \boldsymbol{\theta}_{\mathrm{SRF}}} \left\| \mathbf{Z}_{\mathrm{lr}}^1 - \mathbf{Z}_{\mathrm{lr}}^2 \right\|_1 \tag{8}$$

where $||\cdot||_1$ is a loss function based on the ℓ_1-norm, which describes the discrepancy measure between two LrMSIs. $\boldsymbol{\theta}_{\mathrm{PSF}}^*$

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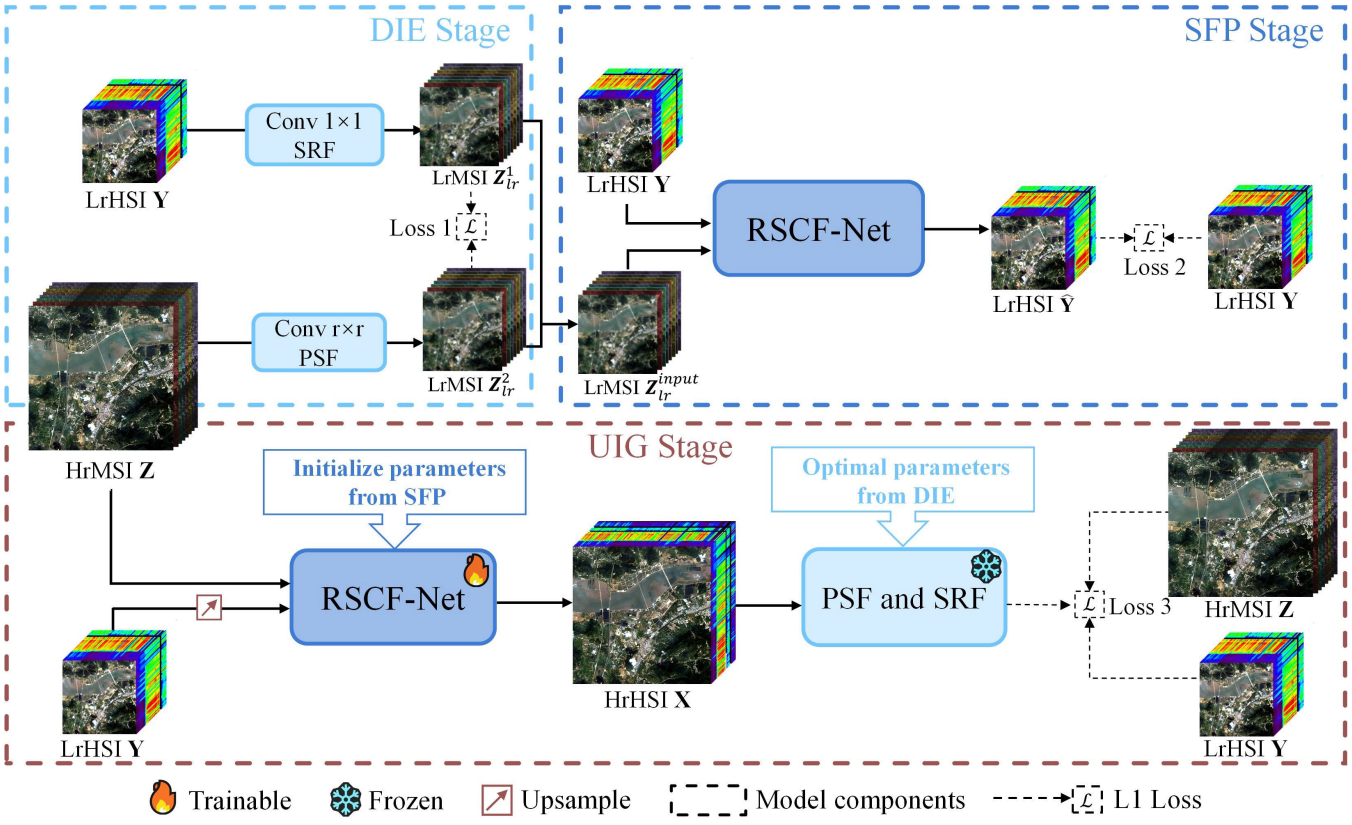


Fig. 2. Overall structures of the proposed MCS-Net. First, the DIE stage is executed, followed by the SFP stage. Finally, the training parameters from these two stages are transferred to the UIG stage.

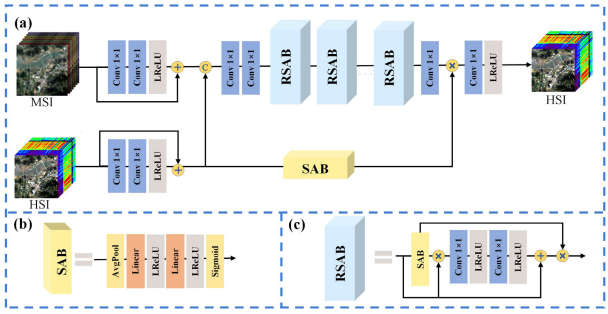


Fig. 3. Detailed structures of (a) RSCF-Net, (b) SAB, and (c) RSAB.

and $\boldsymbol{\theta}_{\text{SRF}}^*$ represent the optimal PSF parameters $\mathbf{P}$ and SRF parameters $\mathbf{S}$ obtained through network optimization, which will be used in the final stage to guide the reconstruction of the unknown HrHSI.

C. Supervised Fusion Pretraining

As mentioned earlier, unsupervised methods are often limited in reconstruction quality due to the lack of accurate reference images. Fortunately, research has shown if correlational relationships between multimodal data can be established from available information and applied effectively, the impact of data scarcity on reconstruction quality can be mitigated [16], [27]. They achieve promising performance by learning the spectral inverse mapping from the available low-resolution data and applying it to the high-resolution data. In contrast, the SFP component is designed to utilize reference images to guide the network in learning the mapping from low-resolution input images to the fusion output process. Specifically, the SFP performs with the support of LrMSI output by DIE, estimating the fusion function and its parameters at low resolution. These parameters are then used to initialize the network in the final stage. For the SFP, we have designed an RSCF-Net based on residual network and attention mechanism, as shown in the architecture of Fig. 3(a).

First, the degraded images of LrMSIs from the two cross-modal datasets are equally important. Therefore, an averaging operation is applied to these images to serve as one of the input data for this process

$$\mathbf{Z}_{\text{lr}}^{\text{input}} = \frac{1}{2}\mathbf{Z}_{\text{lr}}^1 + \frac{1}{2}\mathbf{Z}_{\text{lr}}^2. \tag{9}$$

$\mathbf{Z}_{\text{lr}}^{\text{input}}$ denotes the average operation result of two LrMSIs, and another input is the original LrHSI. The two input images are separately processed for shallow feature extraction using two consecutive 1×1 convolution layers and LeakyReLU activation units. The extracted features are then concatenated along the channel dimension and passed through two additional 1×1 convolution layers for preliminary fusion, adjusting the channel count to a predefined value, P. The fused features are subsequently input into a series of residual spectral attention blocks (RSABs). In each RSAB, the input features first pass

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through a spectral attention block (SAB) based on [59], which performs global average pooling along the spectral dimension to capture spectral global information. This information is then recalibrated via fully connected layers to selectively emphasize useful features, followed by the generation of an attention weight matrix using a Sigmoid function. Subsequently, the attention weights are elementwise multiplied with the input features and added to the feature map obtained by two consecutive convolution layers with LeakyReLU units. Upon output, the residual feature map is again elementwise multiplied with the attention weights to enhance feature representation and facilitate training convergence. The structures of SAB and RSAB are shown in Fig. 3(b) and (c), respectively. Then, the output after N stacked RSABs is passed through a 1×1 convolution layer to adjust the number of channels to match the spectral bands in LrHSI and elementwise multiplied by the attention weights in the initial input features, further reusing the shallow spectral information in LrHSI and correcting the spectral distortions occurring in the fusion branch, thereby enhancing the network's spectral preservation capability. Finally, the features are refined through a convolution layer with a LeakyReLU unit to obtain the final output. The process of modeling the fusion function and its parameters at low resolution through SFP is formalized as follows:

$$\hat{\mathbf{Y}} = \mathcal{F}_r\left(\mathbf{Z}_{\text{lr}}^{\text{input}}, \mathbf{Y}; \boldsymbol{\theta}_{\text{RSCF}}\right) \tag{10}$$

where $\mathcal{F}_r(\cdot)$ represents the RSCF-Net fusion network, and its parameters are denoted as $\boldsymbol{\theta}_{\text{RSCF}}$. The optimization using the power of supervised learning is as follows:

$$\min_{\boldsymbol{\theta}_{\text{RSCF}}} \left\| \mathbf{Y} - \hat{\mathbf{Y}} \right\|_1. \tag{11}$$

Here, $||\cdot||_1$ is used to describe the regression error between the fusion output $\hat{\mathbf{Y}}$ at low resolution and $\mathbf{Y}$.

D. Unsupervised Image Generation

Due to the inherent limitation of supervised methods relying on GT data, image fusion at full resolution is typically performed using unsupervised methods. Although some current methods have reconstructed HrHSI by seeking the mapping relationship from low-resolution to full-resolution images, directly applying this mapping overlooks the data distribution differences caused by the resolution gap. In other words, such methods may lead to significant distortion when handling full-resolution images and exhibit poor model generalization performance. Therefore, we constructed the UIG component to generate the target image unsupervised with the support of the first two components. Specifically, the weights of RSCF-Net trained on low-resolution images are used to initialize the fusion network of UIG, which effectively prevents overfitting. Furthermore, the training in the UIG stage is fine-tuning the pretrained parameters of RSCF-Net, enabling the network to process full-resolution images efficiently and avoid distortion caused by resolution differences. In order to optimize in an unsupervised manner, the PSF and SRF parameters learned by the DIE component are directly used to convert the generated latent target image into its spatial and spectral degraded forms. The fine-tuning process can be formalized as follows:

$$\begin{aligned} &\min_{\boldsymbol{\theta}_{\text{ft}}} \left\| \mathbf{Z} - \mathcal{F}_z\left(\hat{\mathbf{X}}; \boldsymbol{\theta}_{\text{PSF}}^*\right) \right\|_1 + \left\| \mathbf{Y} - \mathcal{F}_y\left(\hat{\mathbf{X}}; \boldsymbol{\theta}_{\text{SRF}}^*\right) \right\|_1 \\ &\text{s.t. } \hat{\mathbf{X}} = \mathcal{F}_r\left(\mathbf{Z}, \mathbf{Y}_{\uparrow}; \boldsymbol{\theta}_{\text{RSCF}}\right) \end{aligned} \tag{12}$$

where $\mathbf{Y}_{\uparrow}$ denotes the upsampling of $\mathbf{Y}$ to match the size of $\mathbf{Z}$ using bilinear interpolation, with the sampling ratio equal to the ratio factor of the input image's GSD. $\hat{\mathbf{X}}$ indicates the generated latent HrHSI in the fine-tuning process. The first term $||\cdot||_1$ indicates the loss between $\hat{\mathbf{X}}$ after spatial degradation and HrMSI, and the second term describes the loss between $\hat{\mathbf{X}}$ after spectral degradation and LrHSI. $\boldsymbol{\theta}_{\text{PSF}}^*$ and $\boldsymbol{\theta}_{\text{SRF}}^*$ are the parameterized representations of the spatial degradation matrix and spectral degradation matrix learned by DIE, respectively, and are fixed during this optimization process. $\boldsymbol{\theta}_{\text{ft}}$ represents the parameters for fine-tuning $\boldsymbol{\theta}_{\text{RSCF}}$ at full resolution.

When the optimal full-resolution fusion parameters $\boldsymbol{\theta}_{\text{ft}}^*$ are obtained, the fusion result is expressed as

$$\mathbf{X} = \mathcal{F}_r\left(\mathbf{Z}, \mathbf{Y}_{\uparrow}; \boldsymbol{\theta}_{\text{ft}}^*\right). \tag{13}$$

$\mathbf{X}$ represents the fusion result HrHSI output by MCS-Net.

IV. Experimental Results

This section provides a comprehensive experimental evaluation of the proposed MCS-Net. First, we detail the datasets and their configurations used in the experiments. Then, a systematic analysis of the ablation experiment results is conducted to validate the effectiveness of each method component. Next, the method's reconstruction performance is thoroughly validated on four simulated datasets and one real satellite dataset. Finally, the compatibility and effectiveness of the proposed semi-supervised framework with existing deep networks are discussed.

A. Experimental Datasets

1) Simulated Datasets: We primarily utilized four hyperspectral remote sensing datasets to conduct simulated experiments.

The Data Fusion Contest (DFC) 2018 Houston dataset [60], captured by the ITRES CASI 1500 camera, consists of 48 bands covering wavelengths from 380 to 1050 nm, with a GSD of 1 m. A subimage of dimensions 400×400 pixels with 46 bands was selected for our experiments.

The Washington DC dataset, acquired in 1995 by the HYDICE sensor, consists of 210 bands spanning wavelengths from 400 to 2500 nm, with a GSD of 2.5 m. For our experiments, we selected a 300×300 pixel subimage from the center of the image. After removing 19 noisy bands, 191 bands remain for use.

The Tiangong-1 dataset contains various representative land cover scenes and spans the visible to shortwave infrared spectrum. In the experiment, an airport image with 54 visible and near-infrared spectral bands, a GSD of 10 m, and a size of 240×240 pixels was selected.

The Pavia University dataset, captured by the ROSIS-3 sensor in 2003, has a GSD of 1.3 m and spans the spectral

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TABLE I
SIMULATED DATASET SPECIFICATIONS

Datasets	Size of HrHSI	Scale factor	Size of LrHSI	Size of HrMSI
Houston	400 × 400 × 46	8	50 × 50 × 46	400 × 400 × 8
Washington DC	300 × 300 × 191	10	30 × 30 × 191	300 × 300 × 8
TianGong-1	240 × 240 × 54	12	20 × 20 × 54	240 × 240 × 8
Pavia University	320 × 320 × 103	16	20 × 20 × 103	320 × 320 × 8

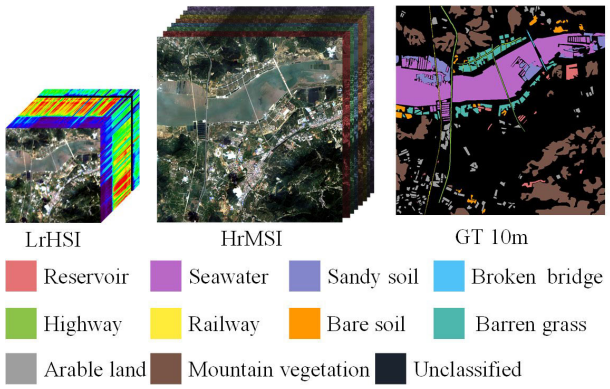


Fig. 4. LrHSI, HrMSI, and the corresponding 10-m GT in the LN01 dataset.

range from 466 to 834 nm with 115 bands. For this study, we selected a subimage from the bottom-left corner with dimensions 320 × 320 pixels and 103 bands, excluding bands with noise and water vapor absorption.

2) Real Dataset: The real satellite data used for classification validation were acquired by the ZY-1 02D satellite in China. Equipped with hyperspectral and multispectral sensors, this satellite can simultaneously capture HSI and MSI of the same scene. The HSI spans 166 bands in the 400–2500-nm range, with a GSD of 30 m. The MSI spans eight bands in the range of 452–1047 nm, with a GSD of 10 m [61].

This study used a real dataset named LN01, collected by the ZY-1 02D satellite over Jinpu New District, Dalian, Liaoning, China. This dataset consists of an HSI of size 300 × 300 × 166 and an MSI of size 900 × 900 × 8. The scene contains ten real land cover types. LN01 data and its corresponding GT are illustrated in Fig. 4.

B. Experimental Details

1) Dataset Preparation: All simulated datasets used for visual and quantitative evaluations were constructed through spatial–spectral downsampling of the original data, which were used as the GT to evaluate fusion quality. Specifically, we conducted simulation experiments following the Wald'sprotocol [62]. The LrHSI was generated by applying Gaussian blurring and $r \times r$ downsampling operators to the original data. The HrMSI was produced using the SRF in WorldView 2. To validate the robustness of the model, we applied different downsampling operators to the four datasets. Table I summarizes the specifications of the simulated data used in the experiments.

For the LN01 real dataset, 19 zero-value bands in the LrHSI were first removed, with the remaining 147 bands for use. The LrHSI and HrMSI were normalized to prepare for the subsequent fusion process.

2) Evaluation Metrics: For the evaluation of fusion quality on simulated datasets, six commonly used metrics were selected, including spectral angle mapper (SAM) [63], peak signal-to-noise ratio (PSNR) [64], relative dimensionless global error in synthesis (ERGAS) [65], structure similarity (SSIM), root-mean-square error (RMSE), and universal image quality index (UIQI) [66], [67]. In terms of visual assessment, SAM heat maps, RMSE heat maps, and mean relative absolute error (MRAE) heat maps were provided to visualize the differences in the quality of the fused images.

For real satellite data, we assessed the fusion quality of real data through classification accuracy due to the lack of validated GT. We selected three evaluation metrics for quantitative assessment, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa).

3) Implementation Details: MCS-Net performs independent optimization in each component rather than adopting a simple end-to-end method, as shown in Algorithm 1. The Adam optimizer trains the proposed model with a linearly decaying learning rate adjustment throughout the training cycle. At three stages, the learning rate is initialized at 0.001, 0.002, and 0.001 and is gradually reduced to zero after 6000, 14 000, and 7000 iterations, respectively. The experiment was conducted on the PyTorch DL framework, utilizing Intel i9-14900KF CPU and an NVIDIA GeForce RTX 4070 Ti GPU.

Algorithm 1 Proposed MCS-Net Algorithm

Input: Observed LrHSI $\mathbf{Y}$ and HrMSI $\mathbf{Z}$
Execution Procedure:
Stage1: Degradation Information Estimation
a) Input $\mathbf{Y}$ and $\mathbf{Z}$;
b) Optimizing (8) with 6000 iterations;
c) Obtain $\mathbf{Z}_{lr}^1$, $\mathbf{Z}_{lr}^2$, and well-trained $\boldsymbol{\theta}_{\text{PSF}}^*$ and $\boldsymbol{\theta}_{\text{SRF}}^*$
Stage2: Supervised Fusion Pre-Training
a) Input $\mathbf{Y}$ and $\mathbf{Z}_{lr}^1$, $\mathbf{Z}_{lr}^2$;
b) Optimizing (11) with 14000 iterations;
c) Obtain $\boldsymbol{\theta}_{\text{RSCF}}$ at low-resolution
Stage3: Unsupervised Image Generate
a) Input $\mathbf{Y}$ and $\mathbf{Z}$;
b) Initialize network parameters using $\boldsymbol{\theta}_{\text{RSCF}}$;
c) Optimizing (12) with 7000 iterations under $\boldsymbol{\theta}_{\text{PSF}}^*$ and $\boldsymbol{\theta}_{\text{SRF}}^*$ to fine-tuning the $\boldsymbol{\theta}_{\text{ft}}$;
d) Obtain fusion result HrHSI $\mathbf{X}$
End Procedure

C. Influence of Parameters

This section explores the effect of the number of RSAB blocks N and the set number of feature channels P in RSCF-Net on the reconstruction quality of our method. The experimental results on the TianGong-1 dataset are presented in Fig. 5. Specifically, the blue bars represent the PSNR values, with higher bars indicating better spatial enhancement performance of the fusion results. The red points represent the SAM values, where lower points indicate better spectral preservation

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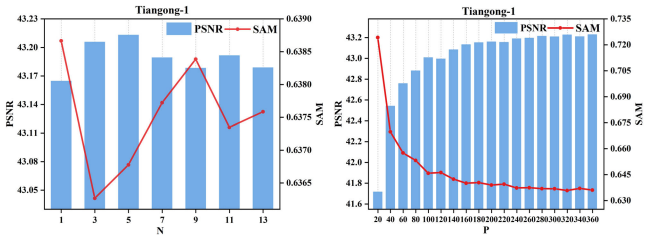


Fig. 5. Relationships between PSNR and SAM in the TianGong-1 dataset under varying block numbers N and feature channel numbers P.

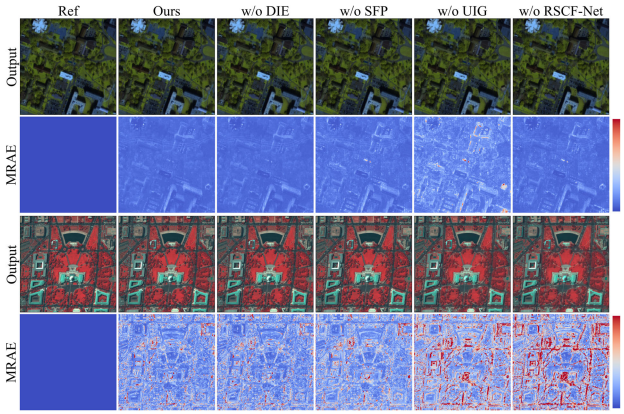


Fig. 6. Ablation study visual results of MCS-Net and its four experimental variants on the Houston and Washington DC datasets. The first and third rows display the pseudo-color images of the fusion results, and the second and fourth rows show the MRAE heat maps.

quality. The left side of the figure shows the relationship between the number of blocks N and SR performance. When N is 3 or 5, SAM and PSNR achieve optimal acceptable values. As N increases, the performance starts to decline. Hence, we set N to 5 to balance these two metrics. The right side of the figure demonstrates the impact of the feature channel number P on SR performance. As P increases, the PSNR and SAM values show significant changes; after P exceeds 240, the trend becomes more gradual. We set P to 280 to balance performance and the number of parameters. Similar trends were observed in experiments on the other three datasets.

D. Ablation Study

In this section, we conduct ablation experiments on the proposed method to assess the contribution of each component. Specifically, we construct four experimental variants by sequentially removing or modifying one of the three main components. These variants allow for a detailed analysis of the impact of each component of MCS-Net on performance. The quantitative results of the experiments are shown in Table II, with the best results highlighted in bold and suboptimal results are shown in blue. The visual comparisons are presented in Fig. 6.

1) Influence of DIE: The DIE component learns the PSF and SRF parameters, which are difficult to obtain in the real world, from the input LrHSI–HrMSI image pairs. This experimental variant aims to investigate the ability of the DIE component to handle the degradation prior to the information acquisition problem and evaluate its impact on the performance of MCS-Net. The corresponding results are shown in Fig. 6 and Table II, labeled as "w/o DIE." In this experiment, we provide accurate PSF and SRF for MCS-Net's first and third stages. The results indicate no clear advantage between the two strategies, regardless of whether the DIE component is used, with the optimal and suboptimal results alternating almost equally between the two strategies. In fact, even in the more complex Washington DC dataset, the unknown response function (i.e., with the DIE component) achieved better PSNR and SAM values. In addition, the blind method always maintains the highest SSIM values in both datasets. These results strongly validate the effectiveness of the DIE component in modeling degradation information.

2) Influence of SFP: We introduced the SFP component to perform SFP on low-resolution images and used the parameters obtained from this training for initialization in the final stage. We designed the "w/o SFP" experimental variant to evaluate its impact, which directly removed the SFP component, allowing MCS-Net to operate entirely unsupervised. The experimental results indicate that the MCS-Net without the SFP component exhibited noticeably poorer quantitative performance and notably lower PSNR values. Moreover, as shown in Fig. 6, it can be observed that the "w/o SFP" variant exhibited more significant detailed distortion in the building and roof regions of both datasets. This phenomenon suggests that introducing supervised pretraining with explicit references and using its pretrained parameters as the initialization strategy for unsupervised fusion can effectively preserve the original data structure, leading to superior reconstruction results. This experiment also confirms the superiority of semi-supervised methods compared to unsupervised methods.

3) Influence of UIG: In the UIG stage, the training parameters of the SFP component are used as the initialization network for the UIG component, followed by fine-tuning using the degradation model parameters from DIE, enabling the network to handle the full-resolution original data effectively. To explore the advantages of this strategy, we designed the "w/o UIG" experimental variant. Specifically, this variant only uses the fusion function and its parameters obtained from training at low resolution to process high-resolution images without further fine-tuning. The quantitative measurement results in Table II show that this variant performs the worst PSNR across all ablation experiments in both datasets. Moreover, compared to the original MCS-Net, other performance metrics also show poorer results. A conclusion consistent with the quantitative measurement results can be drawn from the MRAE heatmap. These results indicate that, as previously mentioned, the training results from low-resolution images may not necessarily apply to high-resolution images. The model fine-tuning strategy significantly improved the network's ability to process full-resolution data, further validating the effectiveness of the UIG component.

4) Influence of RSCF-Net: RSCF-Net extracts potential spatial–spectral information from the input images using a series of modules, enhances feature extraction and reuse through residual structures and squeeze attention mechanisms, and fuses these features. To evaluate the effectiveness of RSCF-Net, we replaced it with a series of simple convolutional

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TABLE II
ABLATION STUDY RESULTS OF MCS-NET ON TWO DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD

Datasets	Methods	SAM	PSNR	ERGAS	RMSE	SSIM	UIQI
Houston	Ours	0.5141	55.9911	0.1505	0.0011	**0.9937**	**0.9999**
	w/o DIE	**0.5128**	**56.0135**	**0.1504**	**0.0010**	0.9933	**0.9999**
	w/o SFP	0.6198	53.9876	0.1919	0.0013	0.9933	0.9998
	w/o UIG	1.0028	48.5363	0.3135	0.0022	0.9876	0.9996
	w/o RSCF-Net	0.5706	55.0817	0.1638	0.0012	0.9936	**0.9999**
Washington DC	Ours	**2.3360**	**43.6226**	0.4479	0.0106	**0.9722**	**0.9985**
	w/o DIE	2.3424	43.5283	**0.4433**	**0.0105**	0.9719	**0.9985**
	w/o SFP	2.3791	40.6959	0.4649	0.0108	0.9581	0.9984
	w/o UIG	3.6592	38.5728	0.6501	0.0155	0.9361	0.9965
	w/o RSCF-Net	4.2206	41.1343	0.7449	0.0178	0.9417	0.9950

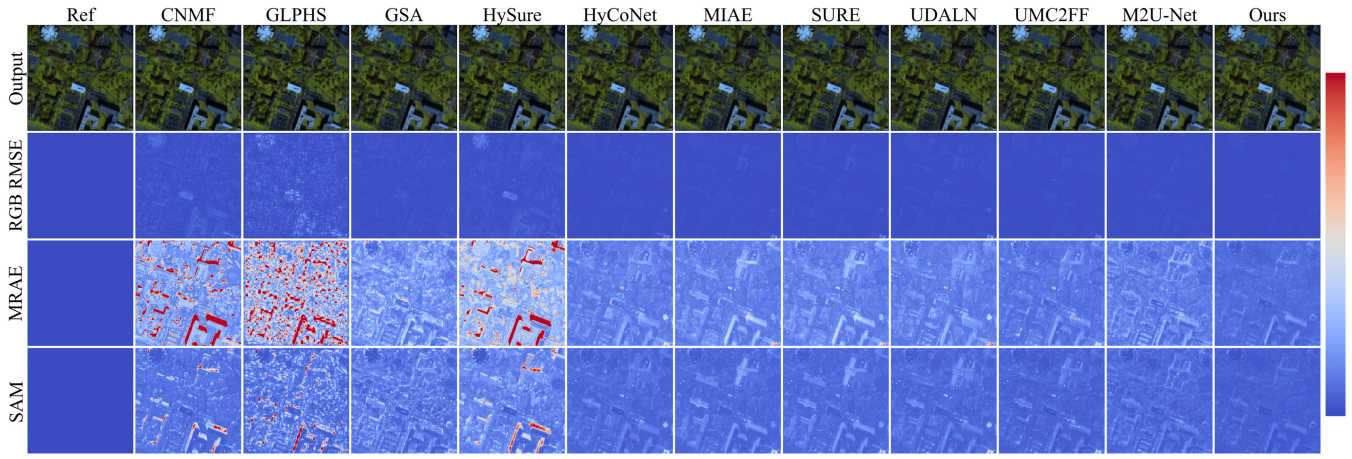


Fig. 7. Visual comparison of fusion results in the Houston dataset. The first row is the color-composite of fusion results; the second row is the RMSE heatmap of color-composite image; the third row is the MRAE heatmap; and the fourth row is the SAM error heatmap.

layers. The experimental results, presented in the "w/o RSCF-Net" measurements in Table II, show that the performance of this variant is slightly inferior to the original MCS-Net on the Houston dataset despite achieving the best and second-best results in the UIQI and SSIM metrics, respectively. However, this variant yielded noticeably poorer performance on the more complex Washington DC dataset, especially in the SAM metric. It demonstrates that RSCF-Net effectively captures spectral information from the data while preserving spatial details, significantly enhancing the model's ability to process data with more complex land cover. This conclusion is further supported by the visual results in Fig. 6.

E. Experimental Results on Simulated Datasets

In this section, we compare our MCS-Net with ten selected state of the arts (SOTAs) from visual and quantitative performance perspectives across four simulated datasets. Among them, GSA [68], GLPHS [35], HySure [36], and CNMF [26] are traditional methods, and HyCoNet [50], MIAE [52], SURE [24], UDALN [16], UMC2FF [27], and M2U-Net [57] are unsupervised DL-based methods. The quantitative and visual results are illustrated in Tables III–VI and Figs. 7–10, respectively.

TABLE III
QUANTITATIVE PERFORMANCE IN THE HOUSTON DATASET

Methods	SAM	PSNR	ERGAS	RMSE	SSIM	UIQI
CNMF	1.2352	41.7493	0.5145	0.0043	0.9856	0.9970
GLPHS	1.7074	38.1425	0.7429	0.0068	0.9778	0.9950
GSA	1.2492	47.1561	0.3312	0.0027	0.9849	0.9995
HySure	1.7342	41.1402	0.6233	0.0045	0.9754	0.9967
HyCoNet	0.7863	49.3494	0.3146	0.0019	0.9860	0.9996
MIAE	0.8540	49.6050	0.3196	0.0019	0.9897	0.9995
SURE	0.8527	49.3609	0.3237	0.0019	0.9884	0.9995
UDALN	0.8011	49.6600	0.3512	0.0019	0.9902	0.9995
UMC2FF	0.6599	52.4574	0.2708	0.0015	0.9900	0.9997
M2U-Net	0.7851	49.7252	0.2513	0.0018	0.9875	0.9998
Ours	**0.5141**	**55.9911**	**0.1505**	**0.0011**	**0.9937**	**0.9999**

1) Visual Performance: The visualized results of all methods across four simulated datasets are shown in Figs. 7–10.

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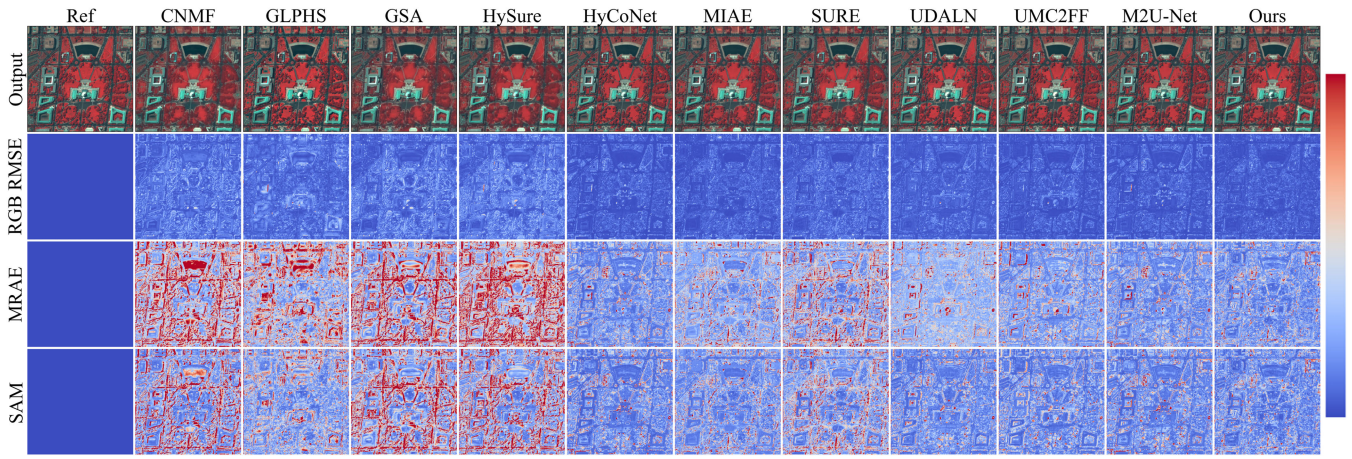


Fig. 8. Visual comparison of fusion results in the Washington DC dataset. The first row is the color composite of fusion results; the second row is the RMSE heatmap of color-composite image; the third row is the MRAE heatmap; and the fourth row is the SAM error heatmap.

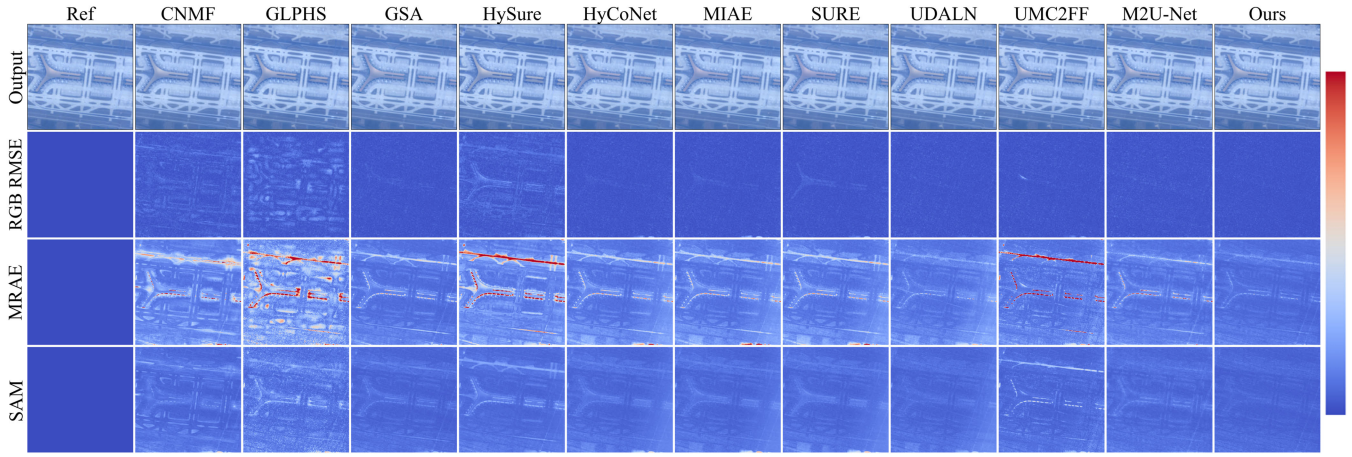


Fig. 9. Visual comparison of fusion results in the TianGong-1 dataset. The first row is the color composite of fusion results; the second row is the RMSE heatmap of color-composite image; the third row is the MRAE heatmap; and the fourth row is the SAM error heatmap.

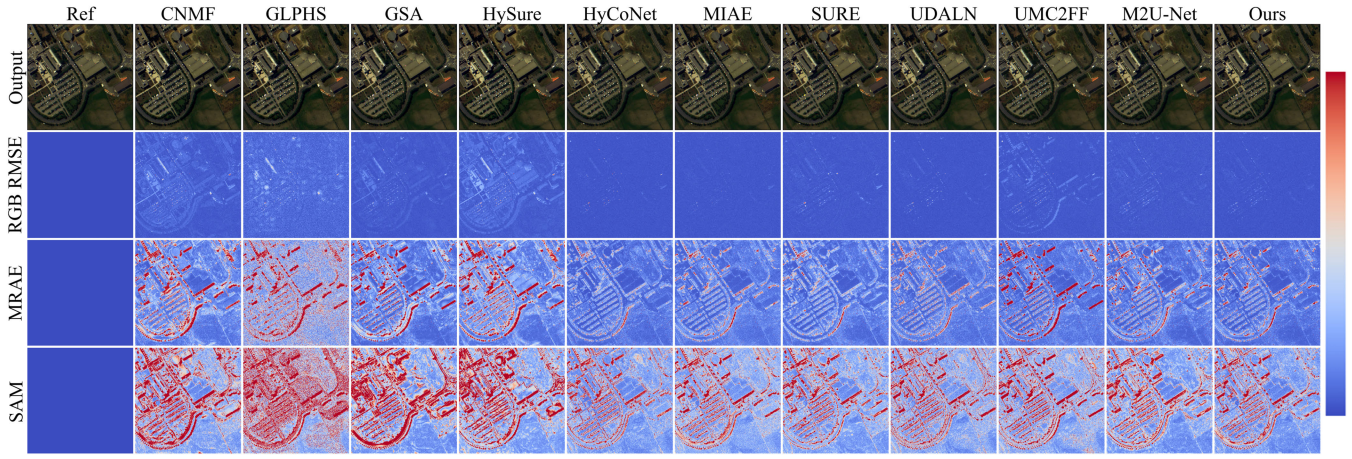


Fig. 10. Visual comparison of fusion results in the Pavia University dataset. The first row is the color composite of fusion results; the second row is the RMSE heatmap of color-composite image; the third row is the MRAE heatmap; and the fourth row is the SAM error heatmap.

The first row shows the SR output's color composite images, the second row displays the RMSE heatmap of the color composite images, the third row presents the MRAE heatmap showing the per-pixel errors of the SR output, and the fourth row illustrates the SAM error heatmap, representing the spectral consistency of each pixel in the SR output. In the visual performance graphs, colors approaching blue denote lower error values. From a visual perspective, the differences between the color composite images of various methods are almost imperceptible to the naked eye, but error heatmaps highlight these discrepancies. Specifically, the CS-based GSA method shows strong spatial preservation among traditional

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TABLE IV
QUANTITATIVE PERFORMANCE IN THE WASHINGTON DC DATASET

Methods	SAM	PSNR	ERGAS	RMSE	SSIM	UIQI
CNMF	4.9524	34.2972	0.8645	0.0192	0.9020	0.9921
GLPHS	3.5733	36.5117	0.7219	0.0168	0.9640	0.9949
GSA	5.3365	39.4743	0.9203	0.0213	0.9136	0.9929
HySure	5.0513	33.6581	0.9284	0.0206	0.8844	0.9927
HyCoNet	2.3580	39.7547	0.5070	0.0116	0.9695	0.9980
MIAE	2.7552	40.9893	0.8489	0.0121	0.9657	0.9964
SURE	3.7545	38.6960	0.7178	0.0162	0.9578	0.9963
UDALN	2.3612	36.7691	0.5324	0.0114	0.9683	0.9977
UMC2FF	2.3551	38.6584	0.4943	0.0108	0.9712	0.9980
M2U-Net	2.4211	38.8236	0.5326	0.0121	0.9689	0.9979
Ours	**2.3360**	**43.6230**	**0.4479**	**0.0106**	**0.9722**	**0.9985**

TABLE V
QUANTITATIVE PERFORMANCE IN THE TIANGONG-1 DATASET

Methods	SAM	PSNR	ERGAS	RMSE	SSIM	UIQI
CNMF	0.9277	37.5087	0.1854	0.0097	0.9502	0.9995
GLPHS	1.1877	34.9847	0.3317	0.0137	0.9290	0.9987
GSA	0.8404	40.2476	0.1713	0.0079	0.9598	0.9995
HySure	0.9258	36.6132	0.2102	0.0105	0.9523	0.9992
HyCoNet	0.8766	39.8675	0.1885	0.0081	0.9601	0.9995
MIAE	0.8772	39.6659	0.1897	0.0082	0.9598	0.9995
SURE	0.8910	39.6263	0.1920	0.0083	0.9594	0.9995
UDALN	0.8474	40.4086	0.1806	0.0080	0.9580	0.9996
UMC2FF	0.8720	39.6773	0.2013	0.0084	0.9529	0.9993
M2U-Net	0.8102	40.6078	0.1338	0.0074	0.9478	0.9998
Ours	**0.6368**	**43.2130**	**0.0946**	**0.0058**	**0.9639**	**0.9999**

TABLE VI
QUANTITATIVE PERFORMANCE IN THE PAVIA UNIVERSITY DATASET

Methods	SAM	PSNR	ERGAS	RMSE	SSIM	UIQI
CNMF	6.3612	31.9743	2.8194	0.0569	0.9287	0.9690
GLPHS	8.5773	29.5690	3.1230	0.0594	0.7754	0.9595
GSA	6.7325	35.9213	2.3418	0.0488	0.7380	0.9726
HySure	8.0760	31.0988	2.9362	0.0584	0.9227	0.9656
HyCoNet	4.2194	37.9948	1.5281	0.0301	0.9031	0.9901
MIAE	4.4735	39.2972	1.5552	0.0300	0.8962	0.9909
SURE	4.4010	38.1594	1.4524	0.0297	0.9140	0.9857
UDALN	5.2121	38.3118	1.7791	0.0364	0.9550	0.9889
UMC2FF	5.5013	34.3281	2.2210	0.0443	0.9489	0.9755
M2U-Net	4.5740	37.1821	1.6069	0.0313	0.8560	0.9902
Ours	**4.1650**	**40.5801**	**1.4373**	**0.0288**	**0.9598**	**0.9919**

methods, but the uneven grayscale distribution between the intensity components of HSI–MSI leads to poor spectral preservation. The matrix decomposition-based CNMF method reconstructs the spatial–spectral matrix with reasonable constraints, and through iterative optimization, it achieves the best results, producing better visual effects. However, traditional methods generally exhibit noticeable spatial and spectral distortions, particularly in images with complex objects. There is a considerable gap when compared to DL-based methods. Most DL-based approaches generate fusion results with high spatial and spectral fidelities. Notably, the UMC2FF method, which reconstructs images from a spectral SR perspective, outperforms other DL methods on the Houston and Washington DC datasets but exhibits noticeable distortions on the other two datasets. Additionally, all DL methods show varying levels of distortion in the edges, complex landform regions, and fine-grained targets of the reconstructed images. For instance, in the TianGong-1 dataset, as shown in Fig. 9, the MRAE and SAM heatmaps reveal reconstruction biases of these DL methods, which are mainly concentrated around the edges of the image and fine-grained targets such as airplanes and bridges, indicated by red and blotchy areas. This phenomenon is primarily caused by the inevitable feature aliasing in low-resolution images, which complicates the reconstruction of complex areas. Our proposed MCS-Net shows the best visual results across four datasets, demonstrating that MCS-Net is robust in reconstructing different scenes and objects and has better SR performance than the competing methods.

2) Quantitative Performance: The quality assessment results of all compared methods across six evaluation metrics are summarized in Tables III–VI, with the best scores highlighted in bold. These results generally align with the findings from visual evaluation. M2U-Net and UDALN show some competitive performance. In Tables III and IV, UMC2FF exhibits lower metrics than M2U-Net and UDALN, indicating its greater sensitivity to data variation. One reason is that it directly applies the learned spectral mapping relationship to generate high-resolution fused images, which limits its generalizability. SURE shows significant spatial–spectral distortion in the Washington DC dataset, while its performance metrics are similar to MRAE in other datasets. In addition, HyCoNet demonstrates relatively stable quantitative results, showing less sensitivity to dataset variations; however, its limited constraints during network learning have restricted its fusion performance. In the traditional methods, GSA produces relatively superior outputs, whereas GLPHS yields the least favorable results. Most importantly, the method we propose demonstrates high robustness to variations in datasets, consistently obtaining the best quantitative measurement scores across all datasets.

Fig. 11 provides the PSNR curves for all methods across each spectral band. Higher curves indicate more substantial spatial reconstruction capabilities, while flatter curves suggest robustness to spectral variation. In the Houston dataset, other methods show a marked PSNR drop in bands 10–15, whereas our method exhibits relatively superior performance in this range. Our model consistently leads in overall trends in the Washington DC and Pavia University datasets. In the TianGong-1 dataset, MCS-Net's PSNR curve remains more stable across all bands, and in the 41st band, where the PSNR values of all methods reach a trough, MCS-Net still maintains a significant advantage. The PSNR curve confirms the conclusions drawn from the quantitative results, further validating the intense competitiveness of our method.

3) Model Parameters and Running Time: We evaluated the running time and model parameter count for MCS-Net and five other DL-based methods, with the results summarized in Table VII. Regarding model parameters, UDALN and UMC2FF have the most minor parameter scales, followed by

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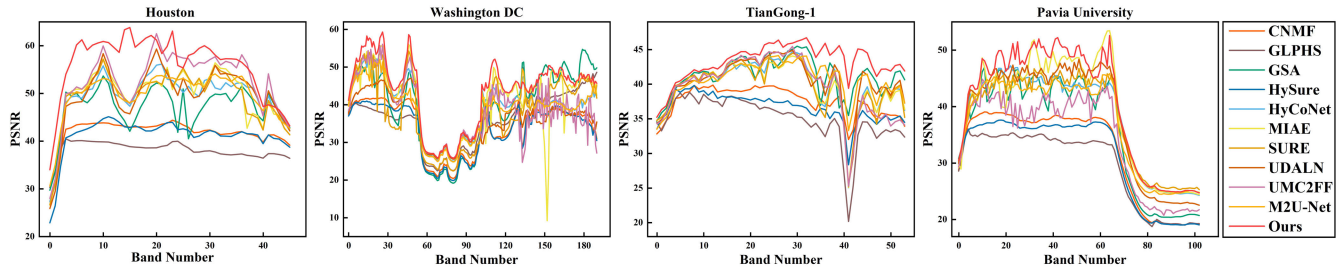


Fig. 11. PSNR value along the spectral band in four datasets.

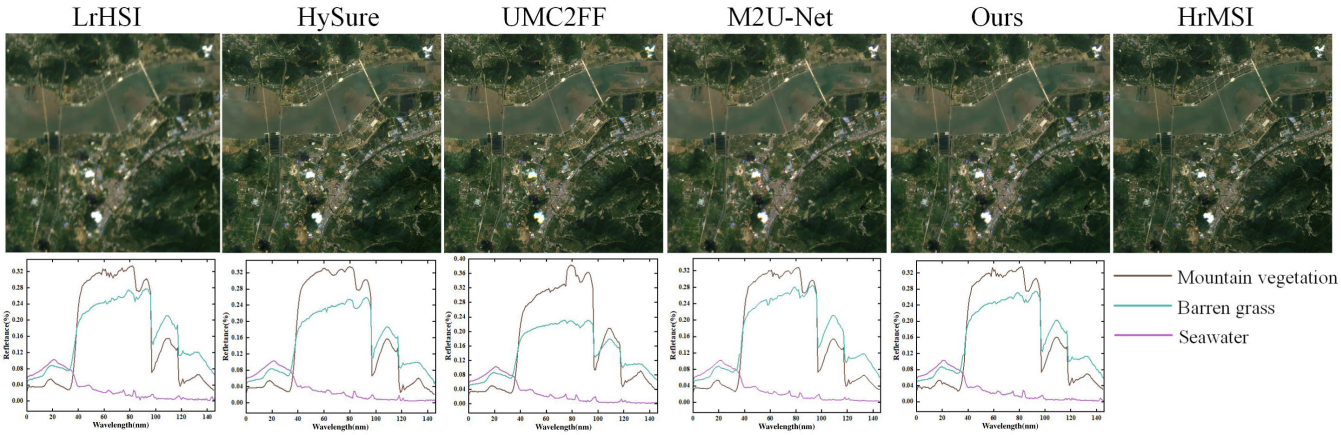


Fig. 12. Visualization of the fusion results for the LN01 dataset. (First row) Visualized results of LrHSI–HrMSI and the HrHSI generated by different SR methods. (Second row) Corresponding visualized spectral curves.

TABLE VII
PARAMETERS AND RUNNING TIME OF DIFFERENT DL-BASED METHODS

Datasets	Methods	Parameters (M)	Running Time (S)
TianGong-1	HyCoNet	2.508	249.92
	MIAE	0.039	208.22
	UDALN	**0.005**	374.39
	UMC2FF	0.008	293.39
	M2U-Net	14.15	838.18
	Ours	1.882	**188.31**
Pavia University	HyCoNet	2.583	860.86
	MIAE	0.141	**240.48**
	UDALN	0.019	765.63
	UMC2FF	**0.012**	763.75
	M2U-Net	14.50	1905.76
	Ours	2.043	533.12

MIAE, the proposed MCS-Net, and HyCoNet, while M2U-Net has the largest number of parameters. The substantial parameter count in M2U-Net is primarily attributed to the dual-U generator network in the final stage and the need to set a large number of feature channels, thus increasing the parameter burden. Interestingly, the running time analysis reveals that despite UDALN and UMC2FF having fewer parameters, MCS-Net demonstrates superior computational efficiency, ranking first and second on the two datasets, respectively. The improvement in efficiency is attributed to the proposed semi-supervised framework, which uses low-resolution images for training in the SFP stage and allocates more optimization iterations for it. Since the image size is smaller, this effectively reduces the overall parameter size and computation time. When processing full-resolution images in the UIG stage, only a few optimization iterations are required to achieve satisfactory reconstruction results.

F. Experimental Results on Real Dataset

In this section, we applied the proposed MCS-Net, the traditional method HySure [36], and the unsupervised methods UMC2FF [27] and M2U-Net [57] to the real satellite dataset LN01, evaluating their fusion results from both visual and quantitative perspectives. In the classification experiments, 10% of each image was selected as training samples. OA, AA, and Kappa were adopted as objective evaluation metrics, and a network with four fully connected layers with dropout (FCN) was employed to analyze the results of LrHSI–HrMSI and the fusion outputs of the four methods. Each image was subjected to ten trials to obtain the final classification accuracy. The visual effects and classification results are shown in Figs. 12 and 13, while the classification accuracy is presented in Table VIII.

1) Visual Performance: Fig. 12 shows the color composite images of LrHSI, HrMSI, and the fusion results (HrHSI) from four methods in the first row for the LN01 dataset, while the second row displays the visualized spectral curves of three land cover types in LrHSI and HrHSI. Compared with the origin LrHSI, all methods generated fusion results containing

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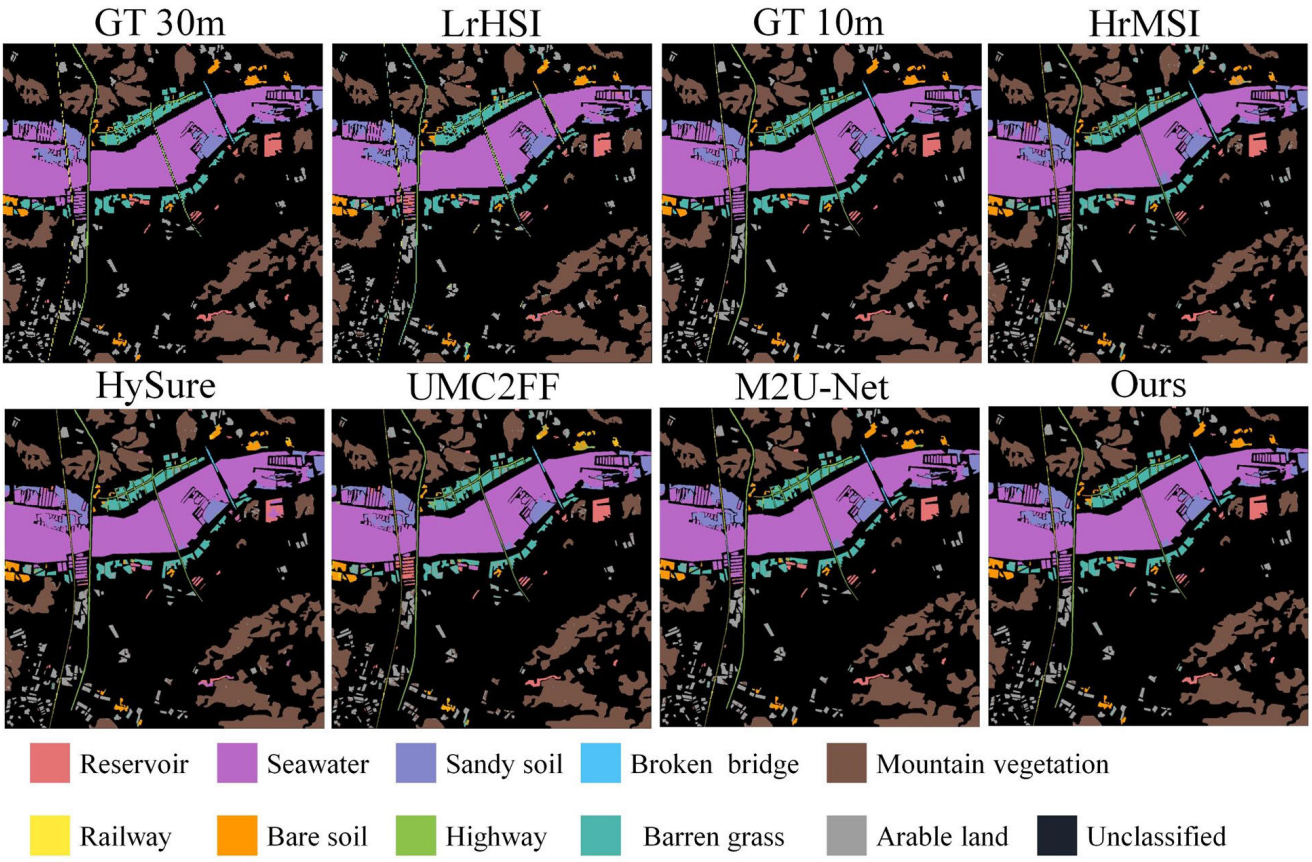


Fig. 13. Classification maps of LrHSI–HrMSI and different SR methods for the LN01 dataset, with overall classification accuracies given in parentheses. LrHSI (93.14 ± 0.44). HrMSI (94.43 ± 0.32). HySure (93.98 ± 0.87). UMC2FF (93.11 ± 0.37). M2U-Net (96.47 ± 0.58). Ours (**96.83** ± **0.26**).

TABLE VIII
QUANTITATIVE CLASSIFICATION PERFORMANCE OF LRHSI–HRMSI AND DIFFERENT SR METHODS ON LN01 DATASET UNDER FCN

Class	LrHSI	HrMSI	HySure	UMC2FF	M2U-Net	Ours
Reservoir	80.20 ± 6.27	91.07 ± 1.03	81.37 ± 4.60	79.99 ± 4.28	92.78 ± 1.04	**93.18 ± 0.87**
Seawater	96.68 ± 0.28	97.59 ± 0.20	96.95 ± 0.26	96.68 ± 0.22	98.12 ± 0.12	**98.20 ± 0.21**
Sandy soil	83.40 ± 1.85	85.44 ± 2.07	84.90 ± 1.60	83.42 ± 0.92	88.81 ± 1.06	**89.66 ± 1.58**
Broken bridge	48.42 ± 13.42	89.56 ± 3.74	**95.23 ± 0.63**	94.80 ± 1.72	84.09 ± 6.88	92.33 ± 5.07
Barren grass	86.09 ± 1.38	86.32 ± 1.10	86.15 ± 1.47	84.94 ± 1.45	92.12 ± 2.25	**93.24 ± 0.53**
Highway	72.09 ± 3.41	89.63 ± 1.18	86.79 ± 2.02	80.00 ± 6.61	**91.58 ± 1.60**	91.17 ± 0.64
Railway	24.72 ± 8.48	71.48 ± 5.46	71.70 ± 4.07	64.56 ± 8.73	85.04 ± 3.73	**87.27 ± 1.57**
Bare soil	86.47 ± 2.09	85.54 ± 1.57	85.54 ± 2.88	80.99 ± 1.71	91.58 ± 2.17	**92.63 ± 0.73**
Mountain vegetation	97.62 ± 0.53	97.72 ± 0.19	97.88 ± 0.90	97.46 ± 0.28	98.81 ± 0.46	**99.00 ± 0.19**
Arable land	86.51 ± 3.00	84.04 ± 0.93	84.70 ± 4.29	81.86 ± 2.08	92.62 ± 2.73	**93.99 ± 1.05**
OA(%)	93.14 ± 0.44	94.43 ± 0.32	93.98 ± 0.87	93.11 ± 0.37	96.47 ± 0.58	**96.83 ± 0.26**
AA(%)	74.19 ± 2.22	80.81 ± 1.33	80.12 ± 1.27	78.32 ± 1.28	83.84 ± 1.21	**85.09 ± 0.82**
Kappa(%)	90.43 ± 0.62	92.22 ± 0.47	91.62 ± 1.19	90.37 ± 0.52	95.07 ± 0.80	**95.58 ± 0.37**

more spatial texture details, indicating their ability to effectively extract and reconstruct spatial information from HrMSI. Visually, no significant color distortion was observed, but the visualized spectral curves more intuitively demonstrated the differences among these methods. The fusion results generated by the traditional method HySure exhibited relatively smooth spectral characteristics, while UMC2FF's spectral curves show significant changes compared to LrHSI. Additionally, the fusion results generated by M2U-Net demonstrate more satisfactory spectral curves compared to the previous two methods. This improvement is partly attributed to M2U-Net's utilization of the potential of DIP and the design of a more powerful generator network, which better captures the spatial–spectral prior of latent HrHSI, thereby helping to extract finer-grained

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spectral features while maintaining spatial details, ultimately yielding higher quality reconstruction results. However, compared to these contrasting methods, our approach generates fusion results with the highest consistency with LrHSI data, as the visualized spectral curves show the smallest deviation. This observation can be attributed to the semi-supervised strategy employed in our method, while the fusion network, which integrates residual structures and attention mechanisms, enables the model to more effectively extract relevant spatial and spectral information, resulting in spectral-accurate and spatially rich fusion results.

Fig. 13 illustrates the classification results of LrHSI, HrMSI, and four other methods applied to the LN01 dataset. GT 30 m and GT 10 m represent the GT images with low and high spatial resolutions, respectively. It can be observed that the low spatial resolution causes LrHSI's classification map to be more blurred. Influenced by the pixel mixing effect, LrHSI shows the lowest classification accuracy for small-scale objects in the scene, such as Railways and Broken bridges. HrMSI lacks sufficient spectral information in high-resolution images and is less sensitive to material differences, resulting in significant misclassification in mixed areas of Mountain vegetation and Arable land. HySure misclassifies a larger portion of the Reservoir as Seawater, while UMC2FF also exhibits misclassification between Mountain vegetation and Arable land. The fusion results generated by M2U-Net and the proposed MCS-Net perform best on the visual classification maps. However, ours MCS-Net produces superior visual results regarding land cover areas such as Bare soil, Barren grass, and Arable land.

2) Quantitative Performance: Table VIII reports the quantitative results of running supervised classification ten times on LrHSI, HrMSI, and the four HrHSIs. Compared to the LrHSI and HrMSI classification results, the fusion images generated by MCS-Net exhibit significant improvements across all land cover categories and objective metrics. This highlights the crucial role of image fusion as a postprocessing technique for integrating complementary information and validates our method's practical applicability. Furthermore, our approach achieved the highest classification accuracy in eight land cover categories and the best results across all three objective metrics compared to the three contrast methods. This indicates that our method can more effectively enhance spatial resolution while maintaining excellent spectral fidelity. By comparing the classification results of HrMSI with those of HySure and UMC2FF, it can be seen that merely introducing additional features does not necessarily improve practical performance. In contrast, fusion results that combine high spatial resolution and excellent spectral fidelity are better suited to meet the demands of downstream tasks.

G. Compatibility Analysis

In this section, we conducted experiments to investigate the compatibility and effectiveness of the proposed multistage semi-supervised framework with existing fusion networks. Specifically, MSDCNN [69] and ResTFNet [70], representing two supervised fusion networks, were utilized to replace the RSCF-Net in the SFP component, resulting in two experimental variants referred to as "MCS + MSDCNN" and "MCS + ResTFNet" in Fig. 14. The data simulation process for the Houston and Tiangong-1 datasets followed Section IV-B, with specifications provided in Table I. Since supervised methods rely on training data, further experimental data configurations were required. For the original supervised method, 80% of the data from the central region of the Houston dataset (size $320 \times 320 \times 46$) were used for testing, while the remaining 20% were used for training after zero-padding the cropped central region. For the Tiangong-1 dataset, 75% of the central region data (size $180 \times 180 \times 54$) were used for testing, and the remaining portion was processed in the same manner for training. The training process involved 10 000 iterations with a learning rate of 0.0001. The corresponding test images were used for the variant experiments, with the learning rates of the three components set to 0.001, 0.0008, and 0.0004, respectively. Other configurations were consistent with the original MCS-Net. Each method is tested three times, and the best result is selected as the experimental outcome.

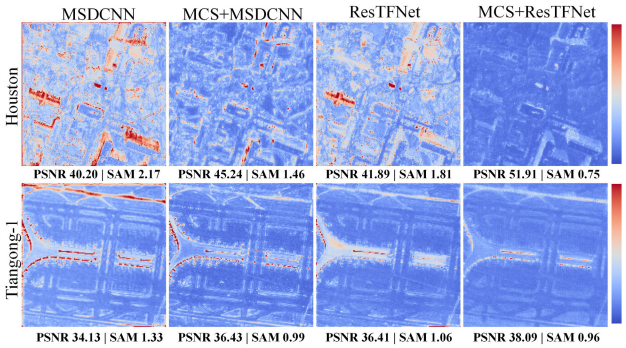


Fig. 14. MRAE heatmap of the fusion results on the Houston and Tiangong-1 datasets for the supervised methods combined with the proposed semi-supervised framework and the original supervised methods.

Fig. 14 displays the MRAE heatmap of the experimental results, offering an intuitive visual representation: even with triplet data for supervised training, MSDCNN's fusion results still exhibit substantial distortions, particularly in building regions. This phenomenon is likely due to the disruption of intrinsic structures caused by multiple downsampling operations during the fusion process. In comparison, ResTFNet shows better performance, yet distortions remain inevitable. Importantly, combining these two methods with the proposed multistage semi-supervised framework results in better outcomes, significantly enhancing areas that the original method cannot effectively handle, such as rooftops and shadows in the Houston dataset and bridges and runway edges in the Tiangong-1 dataset, effectively improving the network's SR performance. The results demonstrate that the proposed multistage semi-supervised framework is flexible and scalable; it can effectively be compatible with most existing deep networks and significantly enhances their fusion performance.

V. Conclusion

This article introduces a semi-supervised network that effectively couples unsupervised learning and supervised pretraining in multistage architecture, MCS-Net for short. The

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proposed method combines the advantages of both to cope with two prominent issues in DL-based HSISR tasks: 1) the impracticality of supervised methods due to their heavy dependence on training triplet datasets and 2) the limited reconstruction quality of unsupervised methods resulting from the lack of reference images and inaccuracies in degradation process estimation. MCS-Net consists of three components with clearly defined functions. First, the DIE extracts deep degradation information from the input LrHSI–HrMSI pair. Then, in the SFP, the supervised learning capability is employed to learn a fusion function at low resolution by the input data pairs and their fused outputs, thereby obtaining pretrained parameters. Finally, the UIG initializes with the pretrained parameter from SFP and fine-tunes the full-resolution fusion network under the guidance of degradation parameters estimated by DIE, enabling efficient handling of full-resolution images. Ablation studies on MCS-Net and its four variants were conducted to evaluate the efficacy of each component alongside a comparative analysis of semi-supervised and unsupervised strategies. On simulated datasets, the proposed method outperformed ten SOTA approaches in terms of both quantitative and visual results. Classification results on real satellite data demonstrated the method's superior practicality. Furthermore, the compatibility experiments with existing deep networks demonstrate the potential for developing the proposed semi-supervised framework. Future research will mitigate overfitting, reduce model parameter size, and investigate how image fusion techniques can handle radiometric differences between LrHSI and HrMSI arising from various factors in practical scenarios.

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